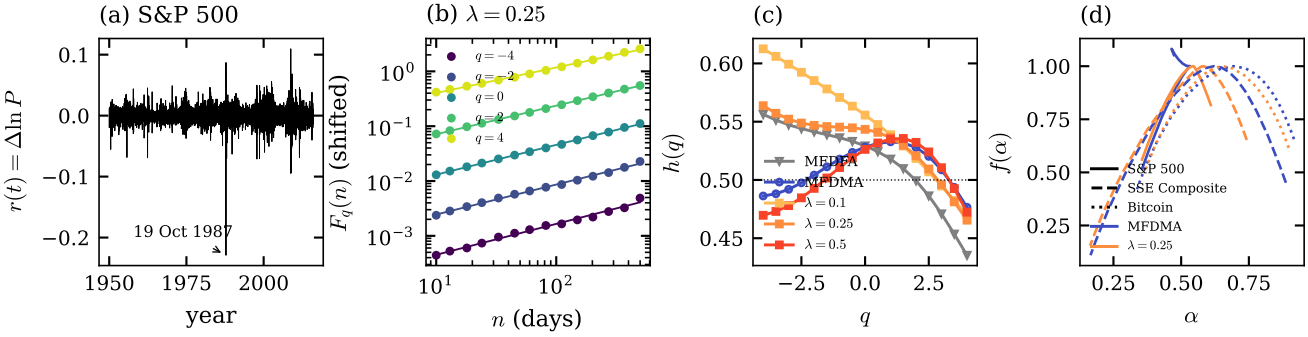

(a) S&P 500
$r(t) = \Delta \ln P$
0.1
0.0
−0.1
−0.2
19 Oct 1987
1950
1975
2000
year
(b) $\lambda = 0.25$
$F_q(n)$ (shifted)
$q = -4$
$q = -2$
$q = 0$
$q = 2$
$q = 4$
10^0
10^{-1}
10^{-2}
10^{-3}
10^1
10^2
n (days)
(c)
$h(q)$
0.60
0.55
0.50
0.45
MFDFA
CMFDMA
$\lambda = 0.1$
$\lambda = 0.25$
$\lambda = 0.5$
−2.5
0.0
2.5
q
(d)
$f(\alpha)$
1.00
0.75
0.50
0.25
S&P 500
SSE Composite
Bitcoin
MFDMA
$\lambda = 0.25$
0.25
0.50
0.75
α